Project Title:

Sales Analytics Dashboard using MySQL, Power BI, and Python

Introduction

This project focuses on analyzing retail sales data using MySQL for database management, Power BI for interactive dashboard creation, and Python for advanced data visualization.

The objective is to transform raw sales data into meaningful business insights through structured analysis and visual storytelling.

Problem Statement

Businesses generate large volumes of sales data but often struggle to extract useful insights.

The goal of this project is to:

- Analyze sales performance
- Identify top-performing regions and products
- Understand revenue and profit trends
- Visualize sales person behavior and sales patterns

Objectives

- Create a structured sales database in MySQL
- Connect MySQL database to Power BI
- Develop interactive dashboards
- Use DAX for KPI calculations
- Implement advanced Python visualizations
- Analyze business performance using multiple chart types

Tools & Technologies Used

Tool	Purpose
MySQL	Database creation & storage
Power BI	Dashboard & visualization
Python	Advanced charts
DAX	Calculations & KPIs

Dataset Description

The dataset represents retail sales transactions.

Key Columns:

- Order ID – Unique order number
- Order Date – Date of transaction
- Region – Sales region
- State – State of sale
- City – City of sale
- Category – Product category
- Sub Category – Product type
- Sales Person – Sales executive
- Quantity – Units sold
- Unit Price – Price per product
- Discount – Discount applied
- Revenue – Total sales amount
- Profit – Profit earned
- Customer Type – New or Returning
- Payment Mode – UPI, Card, Cash

Each row represents a single transaction.

Database Implementation

A MySQL database named powerbi_project was created.

A table sales_analytics was designed with 17 attributes.

200 rows of synthetic sales data were generated for analysis.

Data Connection Process

Steps:

1. Installed MySQL Connector/NET
2. Connected Power BI to MySQL
3. Imported sales_analytics table
4. Performed data validation
5. Created calculated measures using DAX

Visualizations Implemented

Basic Power BI Visuals

- Card (KPIs)
- Bar Chart
- Column Chart
- Line Chart
- Pie Chart
- Donut Chart
- Scatter Plot
- Treemap
- Table
- Matrix
- Funnel Chart
- Gauge Chart

- Waterfall Chart
- Ribbon Chart
- Map & Filled Map

Python Visualizations

- Histogram
- Heatmap
- Pair plot
- Violin Plot
- 3D Scatter Plot
- Network Graph
- Time Series Trend Chart

Python was integrated using Power BI Python scripting.

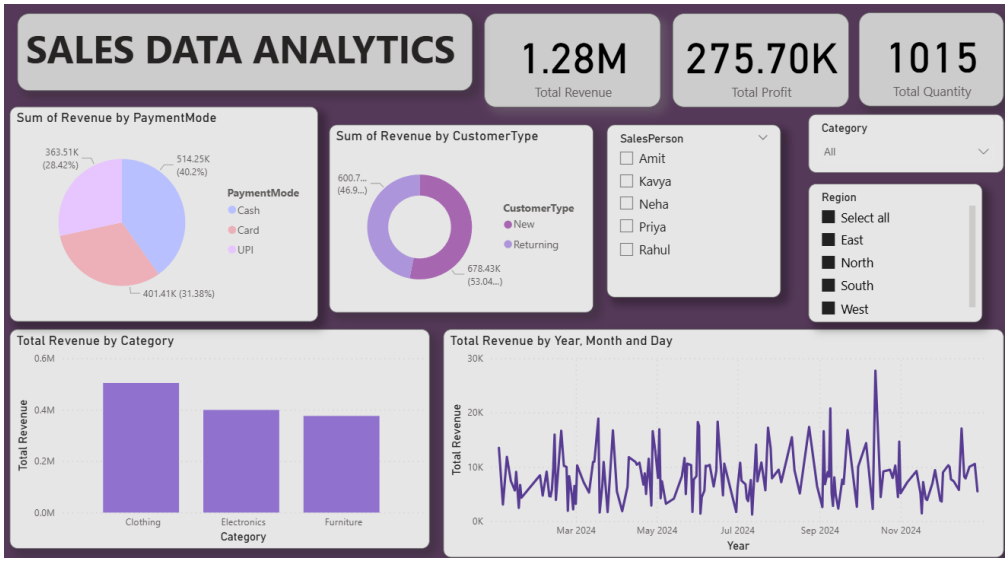
Key Insights

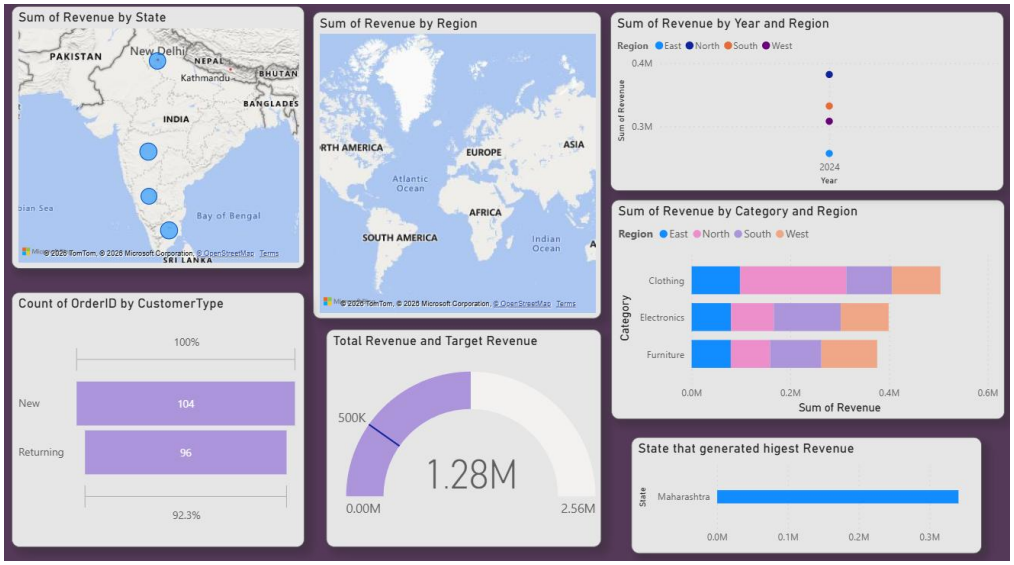
- Electronics & Clothing are top performers
- North region generates highest revenue
- Returning customers are valuable
- Profit strongly depends on revenue
- Maharashtra is top state
- Sales fluctuate monthly
- Cash is preferred payment mode

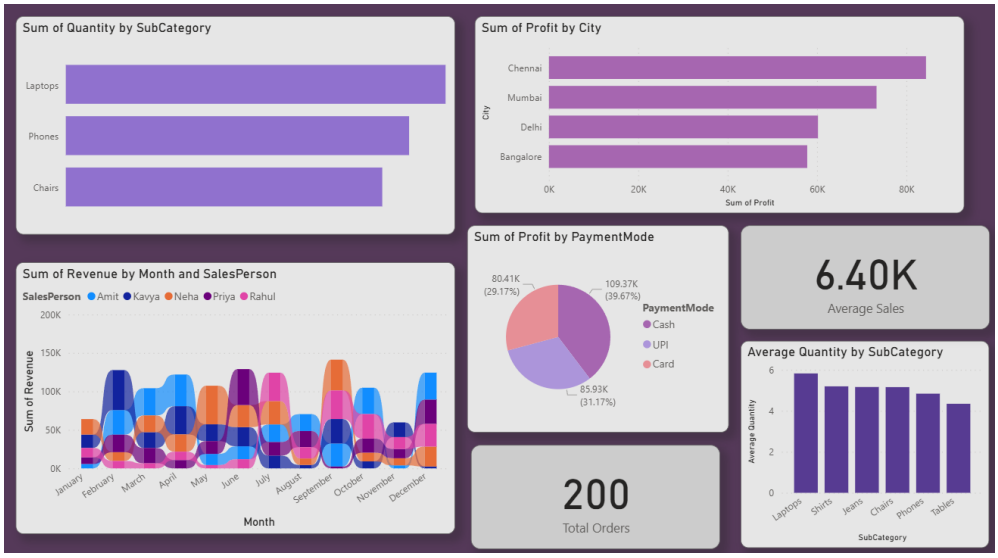
Future Enhancements

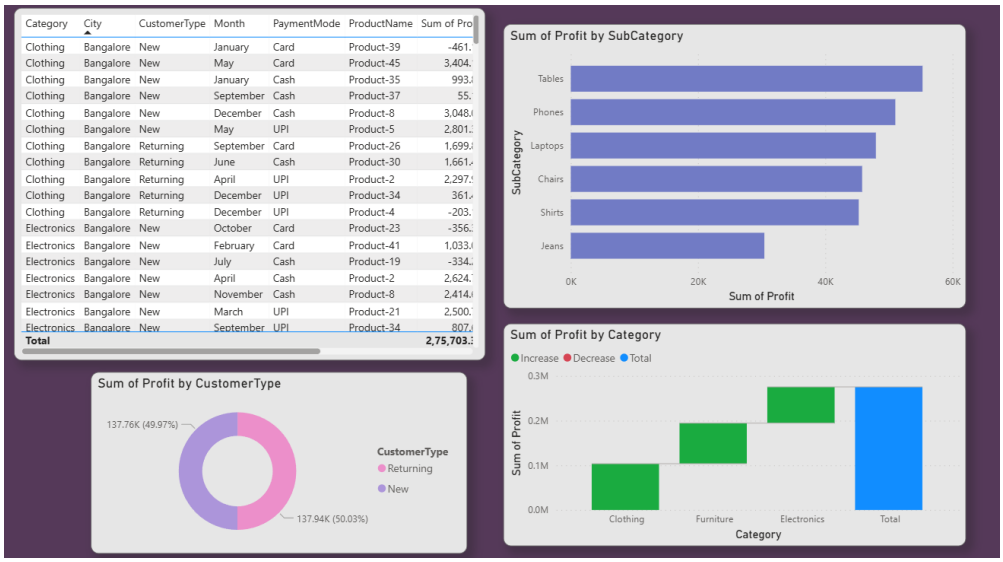
- Real-time data streaming
- Integration with cloud database
- Predictive analytics using Machine Learning
- Customer segmentation analysis

Screenshots

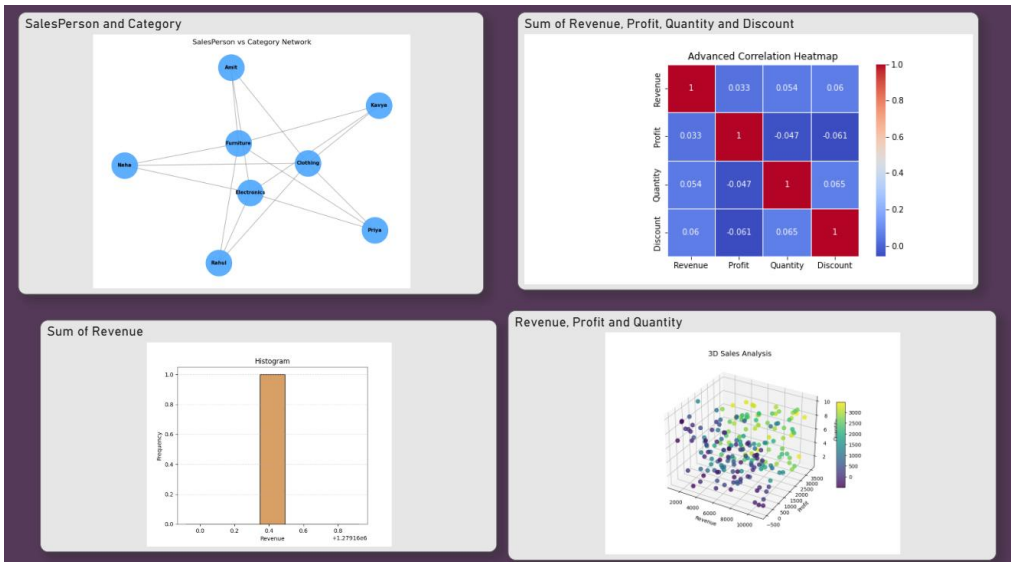


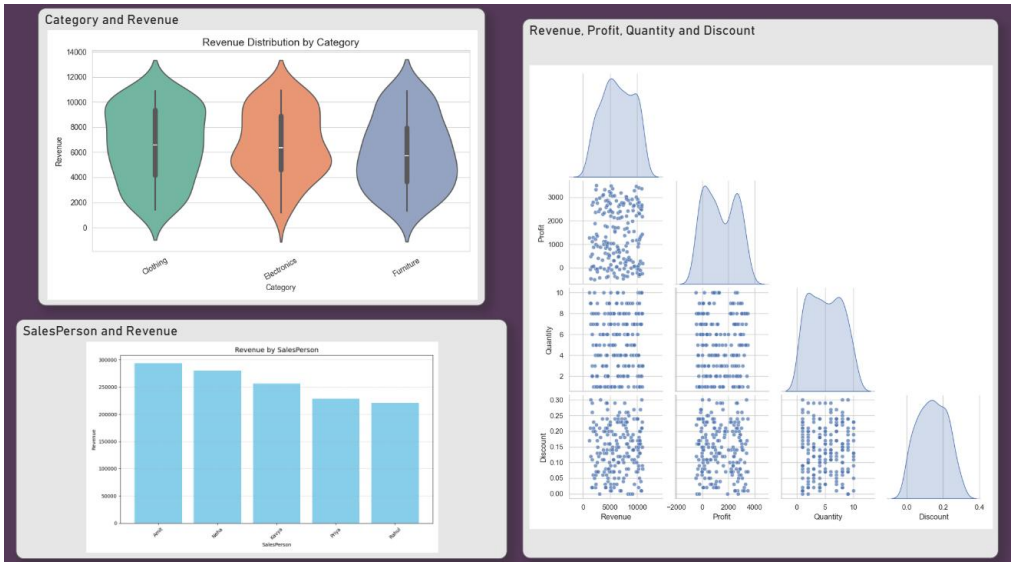






Category	City	CustomerType	Month	PaymentMode	ProductName	Sum of Pro
Clothing	Bangalore	New	January	Card	Product-39	-461.
Clothing	Bangalore	New	May	Card	Product-45	3,404.
Clothing	Bangalore	New	January	Cash	Product-35	993.
Clothing	Bangalore	New	September	Cash	Product-37	55.
Clothing	Bangalore	New	December	Cash	Product-8	3,048.
Clothing	Bangalore	New	May	UPI	Product-5	2,801.
Clothing	Bangalore	Returning	September	Card	Product-26	1,699.
Clothing	Bangalore	Returning	June	Cash	Product-30	1,661.
Clothing	Bangalore	Returning	April	UPI	Product-2	2,297.
Clothing	Bangalore	Returning	December	UPI	Product-34	361.
Clothing	Bangalore	Returning	December	UPI	Product-4	-203.
Electronics	Bangalore	New	October	Card	Product-23	-356.
Electronics	Bangalore	New	February	Card	Product-41	1,033.
Electronics	Bangalore	New	July	Cash	Product-19	-334.
Electronics	Bangalore	New	April	Cash	Product-2	2,624.
Electronics	Bangalore	New	November	Cash	Product-8	2,414.
Electronics	Bangalore	New	March	UPI	Product-21	2,500.
Electronics	Bangalore	New	September	UPI	Product-34	807.
Total						**2,75,703.**





Conclusion

The project successfully demonstrates how sales data can be transformed into meaningful insights using business intelligence tools.

By integrating MySQL, Power BI, and Python, the project showcases advanced data visualization and analytical capabilities.

It provides a comprehensive sales performance dashboard useful for strategic decision-making.